



Decompositions of inequality measures from the perspective of the Shapley–Owen value

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Abstract

This article proposes three new decompositions of inequality measures, drawn from the framework of cooperative game theory. It allows the impact of players' interactions, rather than players' contributions to inequality, to be taken into consideration. These innovative approaches are especially suited for the study of income inequality when the income has a hierarchical structure: the income is composed of several primary sources, with the particularity that each of them is also composed of secondary sources. We revisit the Shapley–Owen value that quantifies the importance of each of these secondary sources in the overall income inequality. Our main contribution is to decompose this importance into two parts: the *pure marginal contribution* of the considered source and a weighted sum of pairwise interactions. We then provide an axiomatic characterization of each additive interaction decomposable (AID) coalitional value considered in this paper. We give an application of these decompositions in the context of inequality theory.

Keywords Game theory · Hierarchical structures · Axiomatic characterization · AID · Inequality measures · Shapley–Owen value

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1 Introduction

In the economic literature, there are two main approaches to decomposing inequality measures. The first one, the decomposition by sub-population, consists of splitting the observed inequality between the groups composing the society. For instance, it may be useful to know whether income inequality is more or less strong within the group of men than it is in the group of women, and to estimate the share of inequality attributed to the income differences between men and women. When this decomposition is considered in the literature, the inequality index often used is the Theil index as its decomposition, first proposed by Theil (1967), allows to split inequality into two components: within-group inequality and between-group inequality. However, there are other measures which are decomposable into sub-populations, such as the Gini index with the decomposition proposed by Dagum (1997), a decomposition that has been extended by Chameni (2006a, b), Mussard et al. (2006), Ebert (2010) and Mornet et al. (2013). The other dominant approach decomposes inequality measures into a set of factors (e.g., income sources). For instance, with a view on public policies, it might be worthwhile evaluating the importance of each income source of income inequality (e.g., labor, capital, social transfers, etc.), especially to appreciate the effectiveness of social transfers in reducing inequality. This decomposition has been notably axiomatized by Shorrocks (1982), but only the decomposition of (the square of) the coefficient of variation respects these axioms. Some authors have departed from these axioms to propose a decomposition of relative inequality measures. For example, Lerman and Yitzhaki (1985) proposed a factor decomposition of the Gini index. Moreover, the multiple decomposition is an approach that takes into account both sub-population effects and factor effects on inequality (e.g., Jenkins & Van Kerm, 2006), and sometimes an additional dimension (e.g., time with Mussard & Savard, 2012 or Mussini, 2013).

Furthermore, all inequality decomposition measures do not use the same methodologies. There are various processes to decompose an inequality index; some are based on mathematical arrangements (all the previously cited authors; with the exception of Lerman & Yitzhaki, 1985), some use statistical tools (e.g., Lerman & Yitzhaki, 1985; Firpo et al. 2018), while others originate from game theory (Chantreuil & Trannoy, 2013; Shorrocks, 2013). In the present article, our innovative decompositions are based on the last category, i.e., cooperative game theory. In the continuity of Auvray and Trannoy (1992), Chantreuil and Trannoy (2013) and Shorrocks (2013), they proposed the Shapley decomposition. This methodology allows to identify and rank the factors that are sources of inequality—in other words, to evaluate the importance of each inequality factor, which is determined by the Shapley value (Shapley, 1953). It should also be noted that this decomposition works with a wide set of inequality measures. Next, this decomposition method was extended by Chantreuil et al. (2019). They pointed out that the importance of a factor in an income inequality does not match with the variation of inequality that would be observed if the income differences associated with it were dropped. This variation corresponds in absolute terms to the pure marginal contribution (PMC) of the source, i.e., the inequality difference between a situation

where all factors are a source of inequality and a situation where one considered factor is no longer a source of differences between individuals. As highlighted by Chantreuil et al. (2019), the importance of the source is equal to its PMC minus a weighted sum of pairwise interactions. We know that a pairwise interaction between two factors is equal to the inequality created by both of them in the presence of another set of factors T , minus the inequality created by each of them in the presence of T , plus the inequality generated by T (see Definition 4). The pairwise interactions highlight the nature of the relation of each pair of factors in the inequality game (complementary, substitutable, and independent) and the intensity of this link. In other words, it indicates to what extent two sources of inequality might favor the same individuals or, on the contrary, to what extent one source can compensate the inequality created by the other. Thus, the inequality variation observed after such a change would be equal to the importance of the considered attribute only if it had no interaction with the other income sources considered in the decomposition. Recently, Courtin et al. (2020) worked on the decomposition of interaction indices. Interaction indices can be viewed as values, which not only assign a payoff to every source but also to every coalition or group of sources (for more information, see Courtin et al. (2020)).

However, the decomposition of the Shapley value proposed by Chantreuil et al. (2019) is not well suited when the income sources have a hierarchical structure. For instance, when the income can be divided into three sources (e.g., capital income, labor income, and social transfers), it is valuable to know the contribution of all “secondary” sources (e.g., financial and estate capital, wage and bonus, family and housing transfers) to their respective “primary” sources to better appreciate the source of inequality. In such cases, Shorrocks (2013) and Chantreuil and Trannoy (2013) proposed to use the Shapley–Owen value (Owen, 1977), notably because the Shapley–Owen value of one primary source is independent from the desegregation of the other primary sources, which is not the case with the Shapley value. Following the previous example, this means that the contribution of labor income to the overall income inequality will not change according to the number of secondary sources considered in the capital income.

Our contribution In this paper, we propose three new decompositions of inequality measures which split into two parts the importance of each source of inequality: its PMC and a term of pairwise interactions (Propositions 5, 6 and 7). However, unlike Chantreuil et al. (2019), we do not decompose an inequality index with the Shapley value but do so more globally using the Shapley–Owen value¹ (Owen, 1977), following the propositions of Shorrocks (2013) and Chantreuil and Trannoy (2013). The Shapley–Owen value belongs to the class of coalitional semivalues introduced and axiomatically characterized by Albizuri and Zarzuelo (2004), while Dubey et al. (1981) defined semivalues. Coalitional semivalues link the classic, well-known coalitional values [Shapley–Owen value (Owen, 1977) and Banzhaf–Owen value (Owen, 1981)], since most of the properties satisfied by the Shapley–Owen value are found to hold also for the Banzhaf–Owen value and most of the coalitional semivalues. From a theoretical point of view, coalitional

¹ Which is well known as a generalization of the Shapley Value.

semivalues can be considered as an alternative or complement to the Shapley–Owen and Banzhaf–Owen values. We propose three decompositions of the Shapley–Owen value and we show that the second decomposition remains valid for the class of coalitional semivalues. Following each decomposition, we provide an axiomatic characterization of additive interaction decomposable (AID) coalitional values. The rest of the paper is a numerical illustration of the proposed methodology, using hypothetical data.

Organization of the paper The next section is mainly devoted to notations and recalls needed to obtain our three decompositions. Three innovative Shapley–Owen decompositions and axiomatic characterizations of AID coalitional values are proposed in Sect. 3, while Sect. 4 provides an application of our decompositions to analyze the inequality of a given hierarchical income distribution. Finally, Sect. 5 concludes the paper, and all the proofs are presented in the [Appendix](#).

2 Preliminaries

Terminology and notation In this paper, sets will be normally represented by uppercase italic letters, while their cardinalities will be denoted by the corresponding lowercase italic letters. We denote by the countably infinite set $\mathcal{U}$, the *universe of all players*, and by $\mathcal{N}$ the set of all finite subsets of $\mathcal{U}$. For any set $E \subseteq \mathcal{U}$, the set of all subsets of E is denoted by 2^E . Any non-empty subset of $\mathcal{U}$ is called *coalition* of $\mathcal{U}$, while any finite partition of $\mathcal{U}$ will be called *coalitional structure* over $\mathcal{U}$. We have determined such a coalitional structure throughout the document and it will be denoted by $\mathfrak{B} := \{B_1, \dots, B_h\}$, $h \geq 1$. Furthermore, for any coalition $T \in \mathcal{N}$, the set $\mathfrak{B}^T := \{X \cap T : X \cap T \neq \emptyset, X \in \mathfrak{B}\}$ is the coalitional structure over T induced by $\mathfrak{B}$.

We will usually consider the finite set of players $N := \{i_1, i_2, \dots, i_n\} \in \mathcal{N}$, $n \geq 1$. Thus, we set $\mathfrak{B}^N := \{S_\ell, \ell \in M\}$ as the coalitional structure over N induced by $\mathfrak{B}$, where $M := \{1, \dots, m\}$ with $1 \leq m \leq n$. Each element of $\mathfrak{B}^N$ will be called a *structural coalition*, and the *representative* in M of any player $i \in N$ is the unique element $m_i \in M$, such that $i \in S_{m_i}$. Consequently, by setting $M_T := \{\ell \in M : S_\ell \cap T \neq \emptyset\}$, for any $T \in 2^N$, $T \neq \emptyset$, the coalitional structure $\mathfrak{B}^T := \{S_\ell \cap T : \ell \in M_T\}$. To avoid heavy notation, for all $T \in 2^N$ and $P \in 2^{M_T}$, we posit $\mathbf{P}_T := \bigcup_{\ell \in P} (S_\ell \cap T)$, in particular $\mathbf{P}_N = \bigcup_{\ell \in P} S_\ell$ and $\mathbf{P}_{N \setminus i} = \bigcup_{\ell \in P} (S_\ell \cap (N \setminus i))$, $i \in N$. $\mathbf{P}_T$ represents the union of all the members of the structural coalitions in P that also are in T .

Let $\pi : \mathcal{U} \rightarrow \mathcal{U}$ be a mapping, and $\pi\mathfrak{B} := \{\pi X : X \in \mathfrak{B}\}$.² Let us define the set $\mathfrak{B}_\pi := \pi\mathfrak{B} \cup \{\mathcal{U} \setminus \pi\mathcal{U}\}$ if $\mathcal{U} \setminus \pi\mathcal{U} \neq \emptyset$, and $\mathfrak{B}_\pi := \pi\mathfrak{B}$ otherwise. Next, $\mathfrak{B}_\pi$ is a coalitional structure over $\mathcal{U}$ if and only if, $\pi B_r \cap \pi B_\ell = \emptyset$ whenever $r \neq \ell$.³

For the sake of convenience, we will omit braces for singletons, e.g., by writing $v(i)$ (resp. $u \setminus i$) instead of $v(\{i\})$ (resp. $u \setminus \{i\}$); similarly for pairs, ij instead of $\{i, j\}$. For $N \in \mathcal{N}$, $L \subseteq N$ and for any $j \in N$, we set

² Notice that, while $\mathfrak{B}$ is a coalitional structure over $\mathcal{U}$, $\pi\mathfrak{B}$ may not be a coalitional structure over $\mathcal{U}$.

³ In that case, $\mathfrak{B}_\pi^N$ is the coalitional structure induced by $\mathfrak{B}_\pi$ over N , and $\pi S_j \cap \pi S_j = \emptyset$ whenever $j \neq k$.

$$\text{Coal}[\mathfrak{B}^L, j] := \{T \subseteq N \mid T := \mathbf{P}_L \cup K; K \subseteq (S_{m_j} \cap L) \setminus j; P \subseteq M_L \setminus \{m_j\}\}. \quad (1)$$

The set K is composed of members of L who are in the same structural coalition as j . The set $\mathbf{P}_L$ is the union of all the members of the structural coalitions over L in $P \subseteq M_L \setminus \{m_j\}$. Then, the union of K and $\mathbf{P}_L$ forms a coalition T . In that case, $\text{Coal}[\mathfrak{B}^L, j]$ represents the set of such T .

2.1 Transferable utility games

Any pair $(\mathfrak{U}, v)$ where v is a function on $2^{\mathfrak{U}}$ satisfying $v(\emptyset) = 0$, and which assigns to every coalition $S \subseteq \mathfrak{U}$ a real number $v(S)$, seen as the worth of S , is called a *TU game*—*Transferable Utility game*. A subset $L \subseteq \mathfrak{U}$ is a carrier of v , if $v(T \cap L) = v(T)$ for all $T \subseteq \mathfrak{U}$. For $N \in \mathcal{N}$, we denote by Γ the set of all games on $\mathfrak{U}$ with finite carriers and by $\Gamma(N)$ the subspace of Γ of games with carrier N . Following usual practice, $(\mathfrak{U}, v)$ will be shortened by simply writing v . In addition, for each $v \in \Gamma$ and any $S \subseteq \mathfrak{U}$, we denote by v^S , the *restricted game* of v upon S , given by $v^S(T) := v(T)$, for all $T \subseteq S$; and we denote by $v_{\cup L}^S$ with $L \subseteq N \setminus S$, the *reduced game* of v upon S in the presence of L , defined by $v_{\cup L}^S(T) := v(L \cup T) - v(T)$, for all $T \subseteq S$. Let $\pi : \mathfrak{U} \rightarrow \mathfrak{U}$ be a mapping, and the pair $(N, \pi v)$ is also a TU game where $\pi v(S) := v(\pi^{-1}S)$ for all $S \subseteq N$. A player $i \in \mathfrak{U}$ is dummy in a game $v \in \Gamma$ if $v(S \cup i) - v(S) = v(i)$ for all $S \subseteq \mathfrak{U} \setminus i$.

Now, given any coalition $T \in 2^{\mathfrak{U}}$; (resp. $T \in 2^N$, $N \in \mathcal{N}$), the *unanimity game* on T denoted by u_T , is defined in such a way that, for all $L \subset \mathfrak{U}$ (resp. $L \subseteq N$), $u_T(L)$ is equal to 1 if $T \subseteq L$, and 0 otherwise. Furthermore, as pointed out by Shapley (1953), the set of unanimity games forms a basis of the space Γ (resp. $\Gamma(N)$). In fact, for any $v \in \Gamma$, $v = \sum_{\substack{T \in 2^{\mathfrak{U}} \\ T \neq \emptyset}} \lambda_T(v) \cdot u_T$, where $\lambda_T(v) := v(T) - \sum_{\substack{S \subsetneq T \\ S \neq \emptyset}} \lambda_S(v)$, for all $T \in 2^{\mathfrak{U}} \setminus \emptyset$ (resp. for any $v \in \Gamma(N)$, $v = \sum_{\substack{T \in 2^N \\ T \neq \emptyset}} \lambda_T(v) \cdot u_T$, where

$$\lambda_T(v) := v(T) - \sum_{\substack{S \subsetneq T \\ S \neq \emptyset}} \lambda_S(v), \text{ for all } T \in 2^N \setminus \emptyset \text{—see Harsanyi dividends (Har-$$

sanyi, 1959).

There is a very rich literature on TU games and their extensions. One of the fundamental problems of TU games is the sharing of the gains from cooperation between the players. A *value* is a function $\varphi : \Gamma \rightarrow \mathbb{R}$ that assigns to every player i in a game $v \in \Gamma$ their prospect $\varphi_i(v)$ of playing the game. Classic values belong to a large family of values called *semivalues*. Semivalues were introduced and axiomatically characterized by Dubey et al. (1981), who also defined a value φ^S as a semivalue when for any $N \in \mathcal{N}$, there exists a family of nonnegative real

Table 1 Values for all coalitions

S	$\{L\}$	$\{T\}$	$\{K\}$	$\{L, T\}$	$\{L, K\}$	$\{T, K\}$	$\{L, T, K\}$
$v(S)$	0.052288	0.026144	0.130719	0.026144	0.183007	0.104575	0.156863

Table 2 The Shapley value of each income source

Income source i	L	T	K	Total
$\varphi_i^{\text{Sh}}(v)$	27.78%	-5.56%	77.78%	100%

numbers $\{q(n, t)\}_{t=0}^{n-1}$, fulfilling $\sum_{t=0}^{n-1} \binom{n-1}{t} q(n, t) = 1$, such that for all $i \in N$ and all $v \in \Gamma(N)$, $\varphi_i^S(v) = \sum_{T \subseteq N \setminus i} q(n, t)[v(T \cup i) - v(T)]$. A semivalue of a player i in a TU game is simply a weighted sum of marginal contributions of this player in the considered game, where the family of weights used is a probability distribution. For instance, the Shapley value (Shapley, 1953), denoted by φ^{Sh} where $q(n, t) = \frac{t!(n-t-1)!}{n!}$, and, the Banzhaf value (Owen, 1975), denoted by φ^{Bz} where $q(n, t) = \frac{1}{2^{n-1}}$, are well-known examples of semivalues.

Let us give an application of semivalues in the context of inequality theory.

Example 2.1 (Inequality game) Let us consider a State where each individual's income comes from three sources: labor (L), social transfers (T), and capital (K). Let us assume that the government's inequality measure is represented by a function v , which measures the inequality produced by the distribution of income sources, such that $v(\emptyset) = 0$. The pair (N, v) is called inequality game⁴ where N is a finite set of income sources. An inequality game is a particular case of TU game applied to inequality theory (Table 1).

The overall inequality is $v(\{L, T, K\}) = 0.156863$. How can we help the decision-maker to significantly reduce income inequality? The decision-maker can use various levers, particularly through labor and capital income taxation and the increase of social transfers. It, therefore, becomes essential for the decision-maker to know the importance of each source. The importance of a source is the contribution of this source to the overall inequality. The importance of a source can be measured from the Shapley value (Table 2).

The capital, social transfers, and labor contribute, respectively, up to 77.78%, -5.56%, and 27.78% to the overall inequality.

2.2 Games with a coalitional structure

A game with a coalitional structure is defined as a triplet $(\mathcal{U}, v, \mathfrak{B})$ where $v \in \Gamma$ and $\mathfrak{B}$ is a coalitional structure over $\mathcal{U}$. Henceforth, we will write $(v, \mathfrak{B})$ instead of the

⁴ For more detail, see Sect. 4.

game $(\mathfrak{U}, v, \mathfrak{B})$. We denote by G the set of all finite games enriched with a coalitional structure and $G(N)$ a subspace of G of games with carrier N . For any $N \in \mathcal{N}$, $M := \{1, 2, \dots, m\}$ is the set of representatives of structural coalitions in $\mathfrak{B}^N$. The *quotient game* $(M, v^q) \in \Gamma(M)$ is the *restricted game* upon M played by the structural coalitions, or rather, by the set M of their representatives as follows: $v^q(P) = v(\mathbf{P}_N)$, for all $P \subseteq M$. This article is mainly based on the concept of *coalitional value*, and by coalitional value, we mean any function Ψ that assigns to every player i in a game $(v, \mathfrak{B}) \in G$ their prospect $\Psi_i(v, \mathfrak{B}) \in \mathbb{R}$ of playing the game. We denote by Ψ^N the restriction of Ψ to $G(N)$.

Among the best-known coalitional values in cooperative games theory, we consider the Shapley–Owen value (Owen, 1977) and the Banzhaf–Owen value (Owen, 1981). We define these two coalitional values as follows:

Definition 1 [*Shapley–Owen value* (Owen, 1977)] For any $N \in \mathcal{N}$ and any $(v, \mathfrak{B}) \in G(N)$, the Shapley–Owen value denoted by Ψ^{Sh} is the coalitional value defined for all $i \in N$ as follows:

$$\Psi_i^{\text{Sh}}(v, \mathfrak{B}) := \sum_{P \subseteq M \setminus m_i} \sum_{K \subseteq S_{m_i} \setminus i} \frac{p!(m-p-1)!}{m!} \frac{k!(s_{m_i}-k-1)!}{s_{m_i}!} [v(\mathbf{P}_N \cup K \cup i) - v(\mathbf{P}_N \cup K)].$$

We recall that $\mathfrak{B}^N := \{B_\ell \cap N : B_\ell \cap N \neq \emptyset, B_\ell \in \mathfrak{B}\} \equiv \{S_\ell : \ell \in M\}$ with $M := \{1, 2, \dots, m\}$.⁵

Definition 2 [*Banzhaf–Owen value* (Owen, 1981)] For any $N \in \mathcal{N}$ and any $(v, \mathfrak{B}) \in G(N)$, the Banzhaf–Owen value denoted by Ψ^{Bz} is the coalitional value defined for all $i \in N$ as follows:

$$\Psi_i^{\text{Bz}}(v, \mathfrak{B}) := \sum_{P \subseteq M \setminus m_i} \sum_{K \subseteq S_{m_i} \setminus i} \frac{1}{2^{m-1}} \frac{1}{2^{s_{m_i}-1}} [v(\mathbf{P}_N \cup K \cup i) - v(\mathbf{P}_N \cup K)],$$

where $\mathfrak{B}^N := \{S_\ell : \ell \in M\}$ with $M := \{1, 2, \dots, m\}$.

The notion of coalitional semivalue was introduced and axiomatically characterized by Albizuri and Zarzuelo (2004).

Definition 3 (*Coalitional semivalue*⁶) Let $\mathfrak{B}$ be a coalitional structure over $\mathfrak{U}$. For any $N \in \mathcal{N}$ and $i \in N$, Ψ^{CS} is a coalitional semivalue, if there are two families of nonnegative real numbers $\{a(m, p)\}_{p=0}^{m-1}$ and $\{b(s_{m_i}, k)\}_{k=0}^{s_{m_i}-1}$ fulfilling the two conditions $\sum_{p=0}^{m-1} \binom{m-1}{p} a(m, p) = 1$ and $\sum_{k=0}^{s_{m_i}-1} \binom{s_{m_i}-1}{k} b(s_{m_i}, k) = 1$, such that

⁵ M being the set of representatives of the structural coalitions in $\mathfrak{B}^N$.

⁶ In the literature, the notion of coalitional semivalue is also known as mixed modified semivalues (see Giménez & Puente, 2015). The semivalue induced by value ϕ plays a role among the structural coalitions whereas the distribution among the players of S_{m_i} is obtained according to the semivalue induced by value φ . Note that the two semivalues can be either equal (e.g., Shapley value for the Owen–Shapley value and Banzhaf value for the Owen–Banzhaf value) or different [e.g., Banzhaf value and Shapley value for the symmetric coalitional Banzhaf value (see Alonso-Meijide & Fiestras-Janeiro, 2002)].

$$\begin{aligned}\Psi_i^{\text{CS}}(v, \mathfrak{B}) &:= \sum_{P \subseteq M \setminus m_i} \sum_{K \subseteq S_{m_i} \setminus i} a(m, p) \cdot b(s_{m_i}, k) [v(\mathbf{P}_N \cup K \cup i) - v(\mathbf{P}_N \cup K)] \\ &= \sum_{T \in \text{Coal}[\mathfrak{B}^N, i]} a(m, p) \cdot b(s_{m_i}, k) [v(T \cup i) - v(T)]\end{aligned}$$

for all $v \in \Gamma(N)$.

The coalitional semivalue of a given player i in a game with a coalitional structure $(v, \mathfrak{B}) \in G(N)$ is equal to the mathematical expectation on $\text{Coal}[\mathfrak{B}^N, i]$ of the marginal contribution $v(T \cup i) - v(T)$ of player i to a coalition $T \in \text{Coal}[\mathfrak{B}^N, i]$ with respect to the probability distribution $\{Q_T\}_{T \in \text{Coal}[\mathfrak{B}^N, i]}$ where $Q_T = a(m, p) \cdot b(s_{m_i}, k)$. We recall that m, p, s_{m_i} and k are determined by T (see Eq. (1)). A coalitional semivalue is simply an extension of a semivalue on a game with coalitional structure.

In particular, the Shapley–Owen value and Banzhaf–Owen value are well-known examples of coalitional semivalues.

Let us propose an application of coalitional semivalues in the context of inequality theory.

Example 2.2 (Example 2.1 continued) In the previous example, we can take our analysis further by studying inequalities at finer levels. We assume that these income sources [labor (L), social transfers (T), and capital (K)] can be subdivided into 2 secondary sources, respectively, wage (L_1) and bonus (L_2); family allowances (T_1) and housing subsidies (T_2); and financial capital (K_1) and estate capital (K_2). Hence, the income is called hierarchical income (Table 3).

The triplet $(N, v, \mathfrak{B}^N)$ is called an inequality game with hierarchical structure where N is a finite set of secondary sources. It is a particular case of a game with a coalitional structure. Let us consider the following table of values (Table 4).

The Shapley–Owen value is a variant of the Shapley value which takes hierarchization into account. The importance of an income source can be measured using the Shapley–Owen value (Table 5).

The distribution of financial capital incomes contributes up to 63.19% to the overall inequality, which is the major contributor to inequality in our example.

We shall consider the following nine properties as *axioms* for a coalitional value Ψ on $(v, \mathfrak{B}) \in G$.

Table 3 Hierarchical structure

	$\mathfrak{B}^N$					
Primary sources (structural coalitions)	L		T		K	
Secondary sources (players)	L_1	L_2	T_1	T_2	K_1	K_2

Table 4 Values for all coalitions

No.	L_1	L_2	T_1	T_2	K_1	K_2	$v(S)$	N°	L_1	L_2	T_1	T_2	K_1	K_2	$v(S)$
1	1	0	0	0	0	0	0.032680	32	1	1	0	0	0	1	0.078431
2	0	1	0	0	0	0	0.019608	33	1	0	1	0	0	1	0.058824
3	0	0	1	0	0	0	0	34	0	1	1	0	0	1	0.045752
4	0	0	0	1	0	0	0.026144	35	1	0	0	1	0	1	0.03268
5	0	0	0	0	1	0	0.104575	36	0	1	0	1	0	1	0.019608
6	0	0	0	0	0	1	0.026144	37	0	0	1	1	0	1	0.013072
7	1	1	0	0	0	0	0.052288	38	1	0	0	0	1	1	0.163399
8	1	0	1	0	0	0	0.03268	39	0	1	0	0	1	1	0.150327
9	0	1	1	0	0	0	0.019608	40	0	0	1	0	1	1	0.130719
10	1	0	0	1	0	0	0.006536	41	0	0	0	1	1	1	0.104575
11	0	1	0	1	0	0	0.006536	42	1	1	1	1	0	0	0.026144
12	0	0	1	1	0	0	0.026144	43	1	1	1	0	1	0	0.156863
13	1	0	0	0	1	0	0.137255	44	1	1	0	1	1	0	0.130719
14	0	1	0	0	1	0	0.124183	45	1	0	1	1	1	0	0.111111
15	0	0	1	0	1	0	0.104575	46	0	1	1	1	1	0	0.098039
16	0	0	0	1	1	0	0.078431	47	1	1	1	0	0	1	0.078431
17	1	0	0	0	0	1	0.058824	48	1	1	0	1	0	1	0.052288
18	0	1	0	0	0	1	0.045752	49	1	0	1	1	0	1	0.03268
19	0	0	1	0	0	1	0.026144	50	0	1	1	1	0	1	0.019608
20	0	0	0	1	0	1	0.013072	51	1	1	0	0	1	1	0.183007
21	0	0	0	0	1	1	0.130719	52	1	0	1	0	1	1	0.163399
22	1	1	1	0	0	0	0.052288	53	0	1	1	0	1	1	0.150327
23	1	1	0	1	0	0	0.026144	54	1	0	0	1	1	1	0.137255
24	1	0	1	1	0	0	0.006536	55	0	1	0	1	1	1	0.124183
25	0	1	1	1	0	0	0.006536	56	0	0	1	1	1	1	0.104575
26	1	1	0	0	1	0	0.156863	57	1	1	1	1	1	0	0.130719
27	1	0	1	0	1	0	0.137255	58	1	1	1	1	0	1	0.052288
28	0	1	1	0	1	0	0.124183	59	1	1	1	0	1	1	0.183007
29	1	0	0	1	1	0	0.111111	60	1	1	0	1	1	1	0.156863
30	0	1	0	1	1	0	0.098039	61	1	0	1	1	1	1	0.137255
31	0	0	1	1	1	0	0.078431	62	0	1	1	1	1	1	0.124183
32	1	1	0	0	0	1	0.078431	63	1	1	1	1	1	1	0.156863

Note to the reader: A “1” means that the source generates inequality, and a “0” means that the income source is equalized (see Sect. 4 for more detail)

Table 5 The Shapley–Owen value of each income source

Income source i	L_1	L_2	T_1	T_2	K_1	K_2	Total
$\Psi_i^{\text{Sh}}(v, \mathfrak{B}^N)$	17.36%	10.42%	0.00%	−5.56%	63.19%	14.58%	100%

- A1. (*Efficiency or carrier* (i)) For all game $(v, \mathfrak{B}) \in G(N)$, $\sum_{i \in N} \Psi_i(v, \mathfrak{B}) = v(N)$.
- A2. (*Carrier* (ii)) For all $(v, \mathfrak{B}), (v, \mathfrak{B}') \in G$, if there exists a carrier N of v , such that $\mathfrak{B}^N = (\mathfrak{B}')^N$, then $\Psi_i(v, \mathfrak{B}) = \Psi_i(v, \mathfrak{B}')$, for all player i .
- A3. (*Linearity*) For all constants $c_1, c_2 \in \mathbb{R}$ and games $(v_1, \mathfrak{B}), (v_2, \mathfrak{B}) \in G$, we have $\Psi_i(c_1 \cdot v_1 + c_2 \cdot v_2, \mathfrak{B}) = c_1 \Psi_i(v_1, \mathfrak{B}) + c_2 \Psi_i(v_2, \mathfrak{B})$, for all player i .
- A4. (*Rearrangement*) For all game $(v, \mathfrak{B}) \in G$, let $\pi : \mathfrak{U} \rightarrow \mathfrak{U}$ be a mapping, such that $\pi B_q \cap \pi B_r = \emptyset$ whenever $q \neq r$. Then, $\Psi_{\pi i}(\pi v, \mathfrak{B}_\pi) = \Psi_i(v, \mathfrak{B})$ for all $i \in B_p$.
- A5. (*Coalitional partnership*) Let $(v, \mathfrak{B}) \in G(N)$, such that B_p is formed by veto players⁷ in v . Let $\pi : \mathfrak{U} \rightarrow \mathfrak{U}$ be a mapping, such that $\pi B_q \cap \pi B_r = \emptyset$, and $|\pi B_p| = 1$. Then, $\Psi_i(v, \mathfrak{B}) = \Psi_i(\Psi_{\pi B_p}(\pi v, \mathfrak{B}_\pi) \cdot u_{B_p}, \mathfrak{B})$, for all $i \in B_p$.
- A6. (*Monotonicity*) Let $(v, \mathfrak{B}) \in G$, such that v is monotonic,⁸ for all $i \in \mathfrak{U}$, we have $\Psi_i(v, \mathfrak{B}) \geq 0$.
- A7. (*Projection*) Let $(v, \mathfrak{B}) \in G$. If v is an additive game,⁹ then $\Psi_i(v, \mathfrak{B}) = v(i)$, for all player i .
- A8. (*Dummy player*) If player i is a dummy in a game $(v, \mathfrak{B}) \in G$, then $\Psi_i(v, \mathfrak{B}) = v(i)$.
- A9. (*Quotient game*) For any $N \in \mathcal{N}$, $(v, \mathfrak{B}) \in G(N)$ and $k \in M$, the following holds, $\sum_{i \in S_k} \Psi_i^N(v, \mathfrak{B}^N) = \Psi_k^M(v^q, [M])$, where $[M] := \{\{1\}, \{2\}, \dots, \{m\}\}$.

Albizuri and Zarzuelo (2004) stated and interpreted the two axioms A4–A5. Moreover, they used the six axioms A2–A7 and provided an axiomatic characterization of coalitional semivalues. The axioms A1–A3 and A6–A9 are standard in the literature (see Alonso-Mejide et al., 2007, etc.). Axiom A8 is stronger than axiom A7. If the mapping π is one-to-one, then A4 becomes the Anonymity axiom.

In the proof of Theorem 7 proposed by Albizuri and Zarzuelo (2004), when we remove the monotonicity and replace the projection by the dummy player property, we obtain the following result:

Theorem 1 *A coalitional value Ψ satisfies carrier (ii), linearity, rearrangement, dummy player property and coalitional partnership, if and only if, for any $N \in \mathcal{N}, (v, \mathfrak{B}) \in G(N)$ and $i \in N$, there exist two families of real numbers*

⁷ A player i is veto in a game v if $v(S) = 0$ whenever $i \notin S$.

⁸ A game v is monotonic if $v(L) \leq v(T)$ whenever $L \subset T$.

⁹ A game $v \in \Gamma(N)$ is an additive game if, for every player $i \in N$, there exists $a_i \in \mathbb{R}$, such that $v(T) = \sum_{i \in N} a_i$ for every $T \subseteq N$.

$\{a(m, p)\}_{p=0}^{m-1}$ and $\{b(s_{m_i}, k)\}_{k=0}^{s_{m_i}-1}$ fulfilling $\sum_{p=0}^{m-1} \binom{m-1}{p} a(m, p) = 1$ and $\sum_{k=0}^{s_{m_i}-1} \binom{s_{m_i}-1}{k} b(s_{m_i}, k) = 1$, such that

$$\Psi_i(v, \mathfrak{B}) := \sum_{P \subseteq M \setminus m_i} \sum_{K \subseteq S_{m_i} \setminus i} a(m, p) \cdot b(s_{m_i}, k) [v(\mathbf{P}_N \cup K \cup i) - v(\mathbf{P}_N \cup K)], \quad (2)$$

where $\mathfrak{B}^N := \{S_\ell : \ell \in M\}$ with $M := \{1, 2, \dots, m\}$.

This result will allow us to give the merits of the different decompositions presented in the next section.

3 Interaction and decomposition

3.1 Notion of interaction

To introduce the notion of interaction considered in this article, we use the approach proposed by Kojadinovic (2005), Grabisch and Roubens (1999) and Fujimoto et al. (2006). A simple way to address this phenomenon consists of considering two players i and j and a characteristic function v , such that if the relation $v(ij) > v(i) + v(j)$ is satisfied, then it seems natural to say that players i and j interact positively. Indeed, considering the pair of players (ij) leads us to a situation in which the value of (ij) is more important (i.e., if two income sources i and j interact positively, the two sources together contribute more to inequality than when they are separated—their interaction increases inequality.). In a symmetrical way, if the relation $v(ij) < v(i) + v(j)$ is satisfied, it seems natural to state that players i and j interact negatively, because considering the pair of attributes (ij) leads us to a situation in which the payoff is lower (i.e., if two income sources i and j interact negatively, the two sources together contribute less to inequality than when they are separated—their interaction decreases inequality). Finally, if the relation $v(ij) = v(i) + v(j)$ is satisfied, we may affirm that players i and j do not interact. Therefore, the interaction between players i and j depends on the difference $v(ij) - v(i) - v(j)$. However, the presence of a third or a fourth player should potentially modify the interaction between players i and j . To take into account this phenomenon, Murofushi and Soneda (1993) introduce what Kojadinovic (2005) calls the interaction between players i and j in the presence of T , where T is a subset of players not containing players i and j . This interaction is given by the following definition.

Definition 4 (*Player pairwise interaction*) For any $N \in \mathcal{N}$, let $v \in \Gamma(N)$ and $i, j \in N$, with $i \neq j$. The interaction between players i and j in the presence of others in $T \subseteq N \setminus ij$ is denoted by $\text{Int}_v^N(i, j, T)$, and defined as follows, $\text{Int}_v^N(i, j, T) := v(ij \cup T) - v(i \cup T) - v(j \cup T) + v(T)$.

Similarly, we define the pairwise interaction between structural coalitions as follows.

Definition 5 (*Structural coalitions pairwise interaction*) For any $N \in \mathcal{N}$, let $(v, \mathfrak{B}) \in G(N)$ and $h, \ell \in M$, with $h \neq \ell$. The interaction between the structural coalitions S_h and S_ℓ in the presence of other structural coalitions in $\mathbf{P}_N := \bigcup_{\ell \in P} S_\ell$ with $P \subseteq M \setminus \{\ell, h\}$, is denoted by $\text{Int}_v^{\mathfrak{B}^N}(S_h, S_\ell, \mathbf{P}_N)$, and defined as follows: $\text{Int}_v^{\mathfrak{B}^N}(S_h, S_\ell, \mathbf{P}_N) := v(S_h \cup S_\ell \cup \mathbf{P}_N) - v(S_h \cup \mathbf{P}_N) - v(S_\ell \cup \mathbf{P}_N) + v(\mathbf{P}_N)$.

Any player i can also interact with each structural coalitional S_ℓ , $\ell \neq m_i$. We define this pairwise interaction as follows:

Definition 6 (*Player and structural coalition pairwise interaction*) For any $N \in \mathcal{N}$, let $(v, \mathfrak{B}) \in G(N)$, $i \in N$, and $\ell \in M \setminus m_i$. The interaction between player i and structural coalition S_ℓ in the presence of others in $T := K \cup \mathbf{P}_N$ with $P \subseteq M \setminus \{\ell, m_i\}$ and $K \subseteq S_m \setminus i$, is denoted by $\overline{\text{Int}}_v^{\mathfrak{B}^N}(i, S_\ell, T)$, and defined as follows: $\overline{\text{Int}}_v^{\mathfrak{B}^N}(i, S_\ell, T) := v(i \cup S_\ell \cup T) - v(S_\ell \cup T) - v(i \cup T) + v(T)$.

Given any $i \in N$, i 's marginal contributions can be represented by the dividends of coalitions they belong to, as presented in the following well-known result.

Proposition 1 For any $N \in \mathcal{N}$ and for all $v, w \in \Gamma(N)$ and $i \in N$, the statements below imply each other.

- (i) For all coalition $T \subseteq N \setminus i$, $v(T \cup i) - v(T) = w(T \cup i) - w(T)$.
- (ii) For all coalition $T \subseteq N \setminus i$, $\lambda_{T \cup i}(v) = \lambda_{T \cup i}(w)$.

A corresponding version of the previous equivalence result, in terms of interactions, is provided in this paper as follows.

Proposition 2 For any $N \in \mathcal{N}$ and for all $v, w \in \Gamma(N)$ and $i, j \in N$, the statements below imply each other.

- (i) For all coalition $T \subseteq N \setminus ij$, $\text{Int}_v^N(i, j, T) = \text{Int}_w^N(i, j, T)$.
- (ii) For all coalition $T \subseteq N \setminus ij$, $\lambda_{T \cup ij}(v) = \lambda_{T \cup ij}(w)$.

3.2 Decompositions of the Shapley–Owen value

In the class of TU games, a natural way to find the payoff of a player i is to look at their effect when they join all other players. Following Casajus and Huettner (2018), this approach is problematic, because one could argue that every player contributes to the payoff of i and, therefore, this contribution should be partly attributed to the others. Chantreuil et al. (2019) proposed a similar approach using the interaction notion.

Following Chantreuil et al. (2019), we propose three approaches for the decomposition of the Shapley–Owen value. Section 3.2.1 is the structural coalition pairwise approach, Section 3.2.2 is the pairwise interaction between a player and a structural coalition approach, and Section 3.2.3 is the player's pairwise interaction approach. We provide an axiomatic characterization of each additive interaction decomposable coalitional value.

One crucial concept in this paper is the *pure marginal contribution*, defined as follows.

Definition 7 [*Pure marginal contribution (PMC)*] For any $N \in \mathcal{N}$ and for all $v \in \Gamma(N)$, the pure marginal contribution PMC_i of a player i is defined by,¹⁰ $\text{PMC}_i(v) := v(N) - v(N \setminus i)$; while the pure marginal contribution PMC_T of a coalition T is defined by,¹¹ $\text{PMC}_T(v) := v(N) - v(N \setminus T)$.

It is worth noting that our pure marginal contributions are not solely defined for a single player, as is done in the literature, but rather for any coalition of N . Now, we can provide our first decomposition of the Shapley–Owen value, as given in the next subsection.

3.2.1 First decomposition approach of the Shapley–Owen value

The notion of AID was used for the first time by Courtin et al. (2020) in the context of interaction indices. We propose a similar definition for the games enriched with a coalitional structure.

Definition 8 [*Basic Additive Interaction Decomposable (AID)*] For any $N \in \mathcal{N}$, let $(v, \mathfrak{B}) \in G(N)$ be a game with coalitional structure. Any coalitional value Ψ is AID if there exists a family of real numbers $\{f(m, p)\}_{p=0}^{m-2}$, such that, for any $S_\ell \in \mathfrak{B}^N$

$$\sum_{i \in S_\ell} \Psi_i(v, \mathfrak{B}) := \text{PMC}_{S_\ell}(v) - \sum_{r \in M \setminus \ell} \sum_{P \subseteq M \setminus \{\ell, r\}} f(m, p) \times \text{Int}_v^{\mathfrak{B}^N}(S_\ell, S_r, \mathbf{P}_N). \quad (3)$$

This definition simply states that, if a coalitional value over $G(N)$ is AID, then the payoff of a given structural coalition expressed as the sum of their members' payoffs should be equal to the difference between the pure marginal contribution of the structural coalition and a weighted sum of pairwise interactions between this structural coalition and the others.¹²

The next result provides an axiomatic characterization of an AID coalitional value.

Theorem 2 (Axiomatization of an AID coalitional value) *A coalitional value Ψ is AID, if and only if, it satisfies the linearity, dummy player property, anonymity and quotient game property.*

Corollary 1 (First decomposition of a coalitional semivalue) *Every coalitional semivalue Ψ^{CS} that satisfies the quotient game property is AID. Furthermore, for any $N \in \mathcal{N}, (v, \mathfrak{B}) \in G(N)$ and $S_\ell \in \mathfrak{B}^N$*

¹⁰ Casajus and Huettner (2018) called the naive solution of player i .

¹¹ Marichal et al. (2007) called the external generalized value of coalition T .

¹² If the family of real numbers $\{f^r(M, P)\}_{P \subseteq M \setminus \{\ell, r\}}$ with $f^r(M, P) = f(m, p)$ is a probability distribution over $2^{M \setminus \{\ell, r\}}$, then the second part of Eq. (3) can be interpreted as a weighted arithmetic mean of pairwise interactions of S_ℓ with the other structural coalitions.

$$\sum_{i \in S_\ell} \Psi_i^{\text{CS}}(v, \mathfrak{B}) := \text{PMC}_{S_\ell}(v) - \sum_{r \in M \setminus \ell} \sum_{P \subseteq M \setminus \{\ell, r\}} f(m, p) \cdot \text{Int}_v^{\mathfrak{B}^N}(S_\ell, S_r, \mathbf{P}_N),$$

where

$$f(m, p) = \begin{cases} \frac{a(m, 0)}{m-1} & \text{if } p = 0 \\ \frac{p \times f(m, p-1) + a(m, p)}{m-p-1} & \text{otherwise,} \end{cases}$$

and the set $\{a(m, p)\}_{p=0, \dots, m-1}$ are nonnegative constants associated to the coalitional semivalue.

From Corollary 1, we can deduce the first decomposition of the Shapley–Owen value.

Corollary 2 (First decomposition of the Shapley–Owen value) *The Shapley–Owen is AID. Furthermore, for any $N \in \mathcal{N}$, $(v, \mathfrak{B}) \in G(N)$ and $S_\ell \in \mathfrak{B}^N$*

$$\sum_{i \in S_\ell} \Psi_i^{\text{Sh}}(v, \mathfrak{B}) := \text{PMC}_{S_\ell}(v) - \sum_{r \in M \setminus \ell} \sum_{P \subseteq M \setminus \{\ell, r\}} \frac{(p+1)!(m-p-2)!}{m!} \times \text{Int}_v^{\mathfrak{B}^N}(S_\ell, S_r, \mathbf{P}_N).$$

3.2.2 Second decomposition approach of the Shapley–Owen value

In the first decomposition of coalitional values, if we remove the quotient game property and replace the anonymity property by rearrangement then, by adding some specific axioms, we obtain another decomposition. Let us first define the second decomposition.

Definition 9 [Additive Interaction Decomposable star (AID $\star$)] For any $N \in \mathcal{N}$, let $(v, \mathfrak{B}) \in G(N)$ be a game and $i \in N$ be a player. A coalitional value Ψ is AID $\star$ if there exist two families of real numbers $\{g(s_{m_i}, k)\}_{k=0}^{s_{m_i}-1}$ and $\{f(m, p)\}_{p=0}^{m-2}$, such that $\sum_{k=0}^{s_{m_i}-1} \binom{s_{m_i}-1}{k} g(s_{m_i}, k) = 1$ and

$$\Psi_i(v, \mathfrak{B}) := \varphi_i(v_i) - \sum_{\ell \in M \setminus m_i} \sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus S_\ell}, i]} f(m, p) \cdot g(s_{m_i}, k) \cdot \overline{\text{Int}}_v^{\mathfrak{B}^N}(i, S_\ell, T),$$

with $v_i = v_{\bigcup(N \setminus S_{m_i})}$ and $\varphi_i(v_i) := \sum_{K \subseteq S_{m_i}} g(s_{m_i}, k) [v_i(i \cup K) - v_i(K)]$.

This definition states that a coalitional value over $G(N)$ is AID $\star$ if the payoff of a given player can be expressed as the difference between a weighted sum of its marginal contributions and a weighted sum of its pairwise interactions with others.¹³

¹³ If we have $\sum_{p=0}^{m-2} \sum_{k=0}^{s_{m_i}-1} \binom{s_{m_i}-1}{k} \times \binom{m-2}{p} \times f(m, p) \cdot g(s_{m_i}, k) = 1$, $g(s_{m_i}, k) \geq 0$ with $k = 0, \dots, s_{m_i}$ and $f(m, p) \geq 0$ with $p = 0, \dots, m-2$, then $\Psi_i(v, \mathfrak{B})$ can also be interpreted as the difference between a weighted sum of marginal contributions of i and a weighted arithmetic mean of pairwise interactions of i with the structural coalitions $S_\ell \neq S_{m_i}$.

The next result provides an axiomatic characterization of $\text{AID}^\star$ coalitional value.

Theorem 3 (Axiomatization of $\text{AID}^\star$ coalitional value) *A coalitional value Ψ is $\text{AID}^\star$, if and only if, it satisfies the linearity, carrier (ii), rearrangement, dummy player property and coalitional partnership.*

Our second decomposition can be applied to any coalitional semivalue as shown below.

Corollary 3 (General decomposition of coalitional semivalue) *Any coalitional semivalue Ψ^{CS} satisfies $\text{AID}^\star$. In addition, for any $N \in \mathcal{N}, (v, \mathfrak{B}) \in G(N)$ and $i \in N$*

$$\Psi_i^{\text{CS}}(v, \mathfrak{B}) = \varphi_i^S(v_i) - \sum_{\ell \in M \setminus m_i} \sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus S_\ell}, i]} f(m, p) \cdot b(s_{m_i}, k) \cdot \overline{\text{Int}}_v^{\mathfrak{B}^N}(i, S_\ell, T),$$

with $\varphi_i^S(v_i) := \sum_{R \subseteq S_{m_i} \setminus i} b(s_{m_i}, r) [v_i(i \cup R) - v_i(R)]$, and

$$f(m, p) := \begin{cases} \frac{a(m, 0)}{m-1} & \text{if } p = 0 \\ \frac{p \times f(m, p-1) + a(m, p)}{(m-p-1)} & \text{if } 0 < p \leq m-2, \end{cases}$$

where $a(m, p)$ and $b(s_{m_i}, k)$ the coefficients associated with Ψ as a coalitional semivalue¹⁴ (p and k are given by $T \in \text{Coal}[\mathfrak{B}^{N \setminus S_\ell}, i]$ see Eq. (1).

Remark 1 Courtin et al. (2020) show that $\varphi_i^S(v_i)$ can be decomposed as follows:

$$\varphi_i^S(v_i) = \text{PMC}_i(v) - \sum_{j \in S_{m_i} \setminus i} \sum_{R \subseteq S_{m_i} \setminus ij} h(s_{m_i}, r) \cdot \text{Int}_{v_i}^N(i, j, R),$$

$$\text{with } v_i = v_{\cup(N \setminus S_{m_i})}^{S_{m_i}}, \text{ and } h(s_{m_i}, r) = \begin{cases} \frac{g(s_{m_i}, 0)}{s_{m_i} - 1} & \text{if } r = 0 \\ \frac{r \times h(s_{m_i}, r-1) + g(s_{m_i}, r)}{(s_{m_i} - r - 1)} & \text{otherwise.} \end{cases}$$

In particular, the second decomposition of the Shapley–Owen value is given by the following result.

Corollary 4 (Second decomposition of the Shapley–Owen value) *For any $N \in \mathcal{N}$, let $(v, \mathfrak{B}) \in G(N)$ be a game N enriched with a coalitional structure, the Shapley value Ψ^{Sh} can be decomposable as follows. For any $i \in N$*

¹⁴ If $\Psi^{\text{CS}} = \Psi^{\text{Sh}}$ or $\Psi^{\text{Sh}} = \Psi^{\text{Bz}}$, then $\sum_{p=0}^{m-2} \sum_{k=0}^{s_{m_i}-1} \binom{s_{m_i}-1}{k} \times \binom{m-2}{p} \times 2f(m, p) \cdot g(s_{m_i}, k) = 1$. $\Psi_i(v, \mathfrak{B})$ can also be interpreted as the difference between a weighted sum of marginal contributions of i and the half of a weighted arithmetic mean of pairwise interactions of i with the structural coalitions $S_\ell \neq S_{m_i}$.

$$\Psi_i^{\text{Sh}}(v, \mathfrak{B}) = \text{PMC}_i(v) - \sum_{j \in S_{m_i} \setminus i} \sum_{R \subseteq S_{m_i} \setminus ij} \text{INT}_{v_i}^{\text{intra}}(i, j, R) \\ - \sum_{\ell \in M \setminus m_i} \sum_{T \in \text{Coal}[\mathfrak{B}^{N_{S_\ell}, i}]} \text{INT}_v^{\text{extra}}(i, S_\ell, T),$$

where $v_i = v_{\cup(N \setminus S_{m_i})}$, $\text{INT}_{v_i}^{\text{intra}}(i, j, R) := \frac{(r+1)!(s_{m_i}-r-2)!}{m!} \text{Int}_{v_i}^N(i, j, R)$ and $\text{INT}_v^{\text{extra}}(i, S_\ell, T) := \frac{(p+1)!(m-p-2)!}{m!} \frac{k!(s_{m_i}-k-1)!}{s_{m_i}!} \overline{\text{Int}}_v^{\mathfrak{B}^N}(i, S_\ell, T)$ (p and k are given by $T \in \text{Coal}[\mathfrak{B}^{N_{S_\ell}, i}]$ see Eq. (1)).

Hence, this second decomposition separates the PMC of each player from their interactions with other players, while differentiating between their pairwise interactions with each player within their own structural coalition and their pairwise interactions with every other structural coalition.

Two structural coalitions can be symmetric to each other with respect to the $\text{AID}^\star$ measure of a coalitional value Ψ . This symmetry is satisfied if the sum of all interactions of each player belonging to a structural coalition S_ℓ with an entire structural coalition S_r , $r \neq \ell$, is equal to the sum of all interactions of each player in S_r with the entire structural coalition S_ℓ . This result is formally given below.

Proposition 3 For any $N \in \mathcal{N}$ and $(v, \mathfrak{B}) \in G(N)$, every structural coalition S_r and S_ℓ are symmetric with respect to the $\text{AID}^\star$ of Ψ^{Sh} , that is

$$\sum_{i \in S_\ell} \sum_{T \in \text{Coal}[\mathfrak{B}^{N_{S_r}, i}]} \text{INT}_v^{\text{extra}}(i, S_r, T) = \sum_{i \in S_r} \sum_{T \in \text{Coal}[\mathfrak{B}^{N_{S_\ell}, i}]} \text{INT}_v^{\text{extra}}(i, S_\ell, T).$$

3.2.3 Third decomposition approach of the Shapley–Owen value

When we maintain the axioms of the second decomposition of coalitional values and add the efficiency property, we obtain another decomposition of the Shapley–Owen value. A coalitional value is $\text{AID}^{\star\star}$, if it is $\text{AID}^\star$ and satisfies the efficiency. The third decomposition imposes more restrictions than the second decomposition. The Shapley–Owen value is the only $\text{AID}^{\star\star}$ coalitional value.¹⁵

Theorem 4 (Third decomposition of the Shapley–Owen value) For any $N \in \mathcal{N}$ and $(v, \mathfrak{B}) \in G(N)$, the Shapley–Owen value is an $\text{AID}^{\star\star}$ coalitional value. For all player $i \in N$

$$\Psi_i^{\text{Sh}}(v, \mathfrak{B}) = \text{PMC}_i(v) - \sum_{j \in N \setminus i} \sum_{T \in \text{Coal}[\mathfrak{B}^{N_{ij}, j}]} \text{INT}_v^{\text{OW}}(i, j, T), \quad (4)$$

where $\text{INT}_v^{\text{OW}}(i, j, T) := Q_T \times \text{Int}_v^N(i, j, T)$, with Q_T given as follows:

¹⁵ Albizuri and Zarzuelo (2004) (Theorem 5) showed that the Shapley–Owen value is the only coalitional value on $(v, \mathfrak{B}) \in G(N)$ that satisfies efficiency, rearrangement, and additivity.

$$Q_T := \begin{cases} \frac{(p+1)!(m-p-2)!}{m!} \times \frac{k!(s_{m_i} - k - 1)!}{s_{m_j}!} & \text{if } S_{m_i} \setminus i = \emptyset \\ \frac{p!(m-p-1)!}{m!} \times \frac{(k+1)!(s_{m_j} - k - 2)!}{s_{m_j}!} & \text{if } S_{m_i} = S_{m_j} \\ \frac{p!(m-p-1)!}{m!} \times \frac{k!(s_{m_j} - k - 1)!}{s_{m_j}!} & \text{if } \emptyset \neq (S_{m_i} \setminus i) \subset T \\ 0 & \text{if } \emptyset \neq (S_{m_i} \setminus i) \subset (N \setminus T) \end{cases}$$

(p and k are given by $T \in \text{Coal}[\mathfrak{B}^{N_i}, j]$ see Eq. 1)).

Moreover, $\sum_{T \in \text{Coal}[\mathfrak{B}^{N_i}, j]} 2Q_T = 1$.

This theorem states that the Shapley–Owen value for a given $i \in N$ in a game with a coalitional structure $(v, \mathfrak{B})$ can be expressed as the difference between its pure marginal contribution $\text{PMC}_i(v)$ and a weighted sum of pairwise interactions between player i and each of the other players in $N \setminus i$.¹⁶

Equation (4) can be rewritten as follows:

$$\begin{aligned} \Psi_i^{\text{Sh}}(v, \mathfrak{B}) &= \text{PMC}_i(v) - \sum_{j \in S_{m_i} \setminus i} \sum_{\substack{T \in \text{Coal}[\mathfrak{B}^N, i] \\ j \notin T}} \frac{p!(m-p-1)!}{m!} \cdot \frac{(k+1)!(s_{m_j} - k - 2)!}{s_{m_j}!} \cdot \text{Int}_v^N(i, j, T) \\ &\quad - \sum_{\ell \in M \setminus m_i} \sum_{j \in S_\ell} \sum_{T \in \text{Coal}[\mathfrak{B}^{N_{S_{m_i}}}, i]} \frac{(p+1)!(m-p-2)!}{m!} \cdot \frac{k!(s_{m_j} - k - 1)!}{s_{m_j}!} \cdot \text{Int}_v^N(i, j, T \cup (S_{m_i} \setminus i)). \end{aligned}$$

Following the definition of $\text{AID}^{\star\star}$, we can also provide the property below.

Proposition 4 For any $N \in \mathcal{N}$ and $(v, \mathfrak{B}) \in G(N)$, two players i and j belonging to the same structural coalition are symmetric with respect to the $\text{AID}^{\star\star}$ of Ψ^{Sh} , that is

$$\sum_{T \in \text{Coal}[\mathfrak{B}^{N_i}, j]} \text{INT}_v^{\text{OW}}(i, j, T) = \sum_{T \in \text{Coal}[\mathfrak{B}^{N_j}, i]} \text{INT}_v^{\text{OW}}(j, i, T).$$

Remark 2 The first, second, and third decompositions of the Shapley–Owen value generalize the main result of Chantreuil et al. (2019).

¹⁶ The second part of the decomposition can also be interpreted as the half of a weighted arithmetic mean of pairwise interactions of i with the other players $j \neq i$.

4 Inequality index decomposition

4.1 An inequality game

This section presents the notion of inequality game and our three decompositions of inequality measures. We first recall that $N := \{i_1, i_2, \dots, i_n\}$ with $n \geq 2$, which will here represent the set of secondary sources that will be defined further. Likewise, $\Omega := \{1, \dots, \omega\}$, with $\omega \geq 2$, will represent the set of individuals, with generic individuals denoted by a, b , while income components are denoted by i, j . We define a situation as an $\omega \times n$ -matrix $x := [x_a^i]$, where $x_a^i \geq 0$ represents the amount of each income i , received by individual a . Thus, row a indicates the amount of income components received by individual a , while column i indicates the distribution of component i between all ω individuals (e.g., Table 6). Let us denote by $\mathcal{X} := \{x | x_a^i \geq 0, \text{ for all } a \in \Omega, i \in N\}$ the set of all situations. With respect to a given situation x , the distribution of income by sources i is given by $x^i := (x_1^i, \dots, x_\omega^i)^\tau$, where τ produces the transposed vector. The aggregate distribution is $X := \sum_{i \in N} x^i$, while the corresponding mean income is given by $\mu(X) := \frac{1}{\omega} \sum_{a \in \Omega} \sum_{i \in N} x_a^i$. Likewise, for all $T \subset N$, the aggregate distribution for the set of sources not included in T is $X^{-T} := \sum_{i \notin T} x^i$, and then, $\mu(X^{-T}) := \frac{1}{\omega} \sum_{a \in \Omega} \sum_{i \notin T} x_a^i$ is the corresponding mean income.

The *inequality index* of a distribution of income X is measured by a function $I: \mathbf{R}^\omega \rightarrow [0, 1]$, such that $I(X) = 0$ for an equal income distribution.¹⁷ Thus, for a given inequality index I , the distribution of income by components helps us to build a distribution of income among subsets of components, namely an application $Y: 2^N \times \mathbf{R}_+ \rightarrow \mathbf{R}^\omega$, such that $Y(\emptyset, \cdot) = 0$ and for all $T \subseteq N$, $Y(T, \lambda) = (\sum_{i \in T} x_1^i + \lambda, \dots, \sum_{i \in T} x_\omega^i + \lambda)$, with $\lambda \in \mathbf{R}_+$. This leads to the definition of a new TU game on N denoted by $v_{I, \lambda}$, and defined by $v_{I, \lambda}(N) := I(X)$ and for all $T \subseteq N$, $v_{I, \lambda}(T) = I(Y(T, \lambda))$ —we called it the λ -inequality game associated with I ; and the parameter λ indicates the way in which the sources not included in any subset T of N must be considered. For instance, they are *nullified* if $\lambda = 0$ or *equalized* if $\lambda = \mu(X^{-T})$.¹⁸

Below are the three decompositions of inequality measures of the paper.

Proposition 5 (1st Decomposition of inequality measures) *Let I be any inequality measure, $\lambda \in \mathbf{R}_+$, $\mathfrak{B}^N$ be a coalitional structure over N and $v_{I, \lambda} \in \Gamma(N)$, with $v_{I, \lambda}$ being the λ -inequality game associated to I . Then, for all situation x*

$$I(X) = \sum_{\ell \in M} \text{PMC}_{S_\ell}(v_{I, \lambda}) - \sum_{\ell \in M} \mathbb{N} \mathbb{T}_{v_{I, \lambda}}^{\text{coalition}}(S_\ell), \quad \text{with } X := \sum_{i \in N} x^i, \quad (5)$$

$$\text{and } \mathbb{N} \mathbb{T}_{v_{I, \lambda}}^{\text{coalition}}(S_\ell) := \sum_{r \in M \setminus \ell} \sum_{P \subseteq M \setminus \{\ell, r\}} \frac{(p+1)!(m-p-2)!}{m!} \times \text{Int}_{v_{I, \lambda}}^{\mathfrak{B}^N}(S_\ell, S_r, \mathbf{P}_N).$$

¹⁷ Other axioms of the inequality measure could be integrated here, such as the strict Schur-convexity, for instance. However, we do not impose more restrictions on the inequality function as our decompositions can be applied with any inequality measure.

¹⁸ See Chantreuil and Trannoy (2011) for the presentation of other games.

Proposition 6 (2nd Decomposition of inequality measures) *Let I be any inequality measure, $\lambda \in R_+$, $\mathfrak{B}^N$ be a coalitional structure over N and $v_{I,\lambda} \in \Gamma(N)$, with $v_{I,\lambda}$ being the λ -inequality game associated to I . Then, for all situation x*

$$I(X) = \sum_{i \in N} \text{PMC}_i(v_{I,\lambda}) - \sum_{i \in N} \text{INT}_{v_{I,\lambda}}^{\text{intra}}(i) - \sum_{i \in N} \text{INT}_{v_{I,\lambda}}^{\text{extra}}(i), \quad (6)$$

with $X := \sum_{i \in N} x^i$ and the functions $\text{INT}_{v_{I,\lambda}}^{\text{intra}}(i) := \sum_{j \in S_{m_i} \setminus i} \sum_{K \subseteq S_{m_i} \setminus \{i,j\}} \text{INT}_{(v_{I,\lambda})_i}^{\text{intra}}(i, j, K)$, and $\text{INT}_{v_{I,\lambda}}^{\text{extra}}(i) := \sum_{\ell \in M \setminus m_i} \sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus \{i\}}]} \text{INT}_{v_{I,\lambda}}^{\text{extra}}(i, S_\ell, T)$.

Proposition 7 (3rd Decomposition of inequality measures) *Let I be any inequality measure, $\lambda \in R_+$, $\mathfrak{B}^N$ be a coalitional structure over N and $v_{I,\lambda} \in \Gamma(N)$, with $v_{I,\lambda}$ being the λ -inequality game associated to I . Then, for all situation x*

$$I(X) = \sum_{i \in N} \text{PMC}_i(v_{I,\lambda}) - \sum_{i \in N} \text{INT}_{v_{I,\lambda}}^{\text{OW}}(i), \quad (7)$$

where $X := \sum_{i \in N} x^i$, and $\text{INT}_{v_{I,\lambda}}^{\text{OW}}(i) := \sum_{j \in N \setminus i} \sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus \{i,j\}}]} \text{INT}_{v_{I,\lambda}}^{\text{OW}}(i, j, T)$.

Proposition 7 states that all inequality measures for a hierarchical income distribution can be decomposed into two terms, the sum of the pure marginal contributions of all income components minus the sum of the sums of all pairwise interactions of all income components with all other income components.

The interpretation of Proposition 6 is different than the interpretation of Proposition 7. In Proposition 7, the interaction is between pairs of income components, while in Proposition 6, the interaction term has two components: a pairwise interaction term between players of the same coalition and a pairwise interaction term between one income component and each coalition of income components. Thus, Proposition 7 is suited to highlight the interactions between each pair of income sources. However, one might suppose that one income source interacts with every other income source only within its structural coalition, and with the other structural coalitions as a whole. In such a case, Proposition 6 should be chosen to make the decomposition as it allows to differentiate between intra- and extra-interactions. Proposition 5 is similar to the Chantreuil et al. (2019) decomposition, with the difference that we do not consider the contribution of one player (with its PMC and its pairwise interaction term), but rather the contribution of one structural coalition (with the structural coalition PMC and the structural coalitions pairwise interaction).

One may notice that the decompositions of the inequality index given by formulae (5), (6), and (7) are similar to the decomposition proposed by Chantreuil et al. (2019), either in the case where the coalitional structure is composed solely of subsets containing a single player (i.e., $m = n$) or when the coalitional structure is only composed of one set, which is actually the whole set N containing all the players (i.e., $m = 1$, while $n \geq 2$).

Furthermore, unlike Chantreuil et al. (2019), and for the sake of clarity, we only consider the equalized inequality game in the next subsection, where the inequality coming from the sources not included in T is equalized. Therefore, we only take into

account the case where $\lambda = \mu(X^{-T})$. Thus, the sources not included in the considered coalition T are equalized so as to drop income differences for these sources. This choice has two main advantages. First, we keep the same amount of total income distributed in the considered economy in all the coalitions. This way, differences in inequality between coalitions do not come from an “income size effect”. Second, the interpretation is more straightforward. Indeed, the contribution to the inequality of an equalizing source is null, and if a source increases the total income inequality, it should have a positive contribution, and vice versa.

Using axiomatic methods, we show that the third decomposition imposes more restrictions than the second decomposition. The third decomposition is the best for the decompositions of inequality with hierarchical structure, because the purpose of a hierarchical structure is to go to an even finer level (primary source). The last decomposition isolates the pure marginal contribution to pairwise interactions between primary sources.

4.2 Decompositions of a hierarchical income distribution

A hierarchical income distribution is defined (Table 6) to illustrate the revisited Shapley–Owen decomposition of an inequality index. Three primary income sources are considered: labor (L), social transfers (T), and capital (K). Each of these sources is subdivided into two secondary sources, respectively: wage (L_1) and bonus (L_2), family allowances (T_1) and housing subsidies (T_2), financial capital (K_1), and estate capital (K_2). We also observe in Table 6 that the total income of individual 3 is higher than the total income of individual 2, itself higher than individual 1’s income. All secondary sources lead to an individual ranking by income, with the exception of family allowances, which are lump-sum transfers, and housing subsidies, which are higher for the poorer people.¹⁹

First, to highlight the interest of using the Shapley–Owen value rather than the Shapley value, we compare the Gini decomposition into income sources with both values between two cases. In the first one, we decompose the inequality measure according to the hierarchical income distribution as given in Table 6. In the second one, we do the same exercise with the difference that the two distributions associated with capital incomes are merged. In Table 7, the contributions of the secondary sources not associated to capital incomes are unchanged when using the Shapley–Owen values, as is also the case of the contribution of the capital income (0.122 in each case). However, with the Shapley value, the change in aggregation of the capital income modifies all contributions (not only its own contribution; more surprisingly, even the ones that did not belong to the capital income source changed).²⁰

¹⁹ Note that all considered sources have nonnegative values. If this is not the case then, one must apply transformations to the distribution to have nonnegative values or take an inequality measure suitable for such a situation, such as the Gini coefficient developed by Raffinetti et al. (2015).

²⁰ These observations are also verified with a null game, where $\lambda = 0$.

Table 6 Hierarchical income distribution

Individual	Labor		Social transfers		Capital		Total
	Wage	Bonus	Family	Housing	Financial	Estate	
1	20	10	10	20	20	30	110
2	35	15	10	10	50	50	170
3	45	25	10	0	100	50	230
Total	100	50	30	30	170	130	510

Table 7 Shapley–Owen and Shapley values sensitivity to aggregation

Value	Case	Labor income		Social transfers		Capital income		
		Wage	Bonus	Family	Housing	Financial	Estate	Sum
Shapley	N° 1	0.02920	0.01720	–	– 0.01440	0.10110	0.02370	0.12480
	N° 2	0.02723	0.01634	–	– 0.01198	0.12527		0.12527
Shapley–Owen	N° 1	0.02720	0.01630	–	– 0.00870	0.09910	0.02290	0.12200
	N° 2	0.02720	0.01630	–	– 0.00870	0.12200		0.12200

The inequality measure is the Gini index. The “Sum” column corresponds to the sum of the financial and estate capital income contributions, in case they are divided. “–” indicates a perfect null value

4.2.1 First decomposition

For brevity’s sake, the first decomposition based on Proposition 5 is not given. Nevertheless, it highlights the possibility to consider structural coalitions as players²¹ when studying the sources of inequality.

4.2.2 Second decomposition

In Table 8, a revisited Shapley–Owen decomposition (based on proposition 6) is proposed to point out the difference between the PMC of an attribute (indicating the variation of inequality that would be observed if the inequality it creates were to be disregarded) and two terms of pairwise interactions (indicating the nature and the intensity of the relationship between this source and the other sources of its structural coalition or any other structural coalition). There are three kinds of relationships: negative, positive, and null. In the first case, a negative relationship means that both sources generate, on average, more inequality together than separately. On the other hand, a positive relationship indicates that both sources create, on average, less inequality together than separately. In specific cases, such as a lump-sum source in an equalized inequality game with the *Gini index* as a

²¹ We consider the Quotient game where structural coalitions are viewed as players (Alonso-Mejide et al., 2007).

Table 8 A revisited Shapley–Owen decomposition—second decomposition

	Labor income		Social transfers		Capital income		Sum
	Wage	Bonus	Family	Housing	Financial	Estate	
Ψ^{Sh}	0.0272	0.0163	–	– 0.0087	0.0991	0.0229	0.1569
	17.36%	10.42%	0.00%	– 5.56%	63.19%	14.58%	100%
PMC	0.0327	0.0196	–	– 0.0261	0.1046	0.0261	0.1569
	20.83%	12.50%	0.00%	– 16.67%	66.67%	16.67%	100.00%
$\text{INT}_{v_{l,j}}^{\text{intra}}$	– 0.0000	– 0.0000	–	–	0.0000	0.0000	0.0000
	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
$\text{INT}_{v_{l,j}}^{\text{extra}}$	0.0054	0.0033	–	– 0.0174	0.0054	0.0033	– 0.0000
	3.47%	2.08%	0.00%	– 11.11%	3.47%	2.08%	– 0.00%

All percentages are calculated relatively to the total inequality. Besides, “–” indicates a perfect null value, while “0.0000” indicates a value close to zero

characteristic function, a source may be independent of others. Results are given in Table 8, where the inequality is measured by the Gini index.²²

According to Table 8, the financial capital income is the most important source of inequality. Moreover, if the financial capital income were no longer a source of inequality, the inequality would decrease by 66.67%, which makes this source the one with the greatest impact on inequality. In addition, as expected, the housing subsidies have a negative effect on inequality. If this source were equalized, then the inequality would increase by 16.67%.

Next, it appears that the pure marginal contributions are higher than their corresponding contributions in absolute terms. Thus, this indicates that removing inequality from those income sources (except the family social transfer source) would reduce (in case of the housing source) the inequality or increase it to an higher level than their corresponding contributions. This also indicates that the sum of interactions has the same sign as the contribution for each income source. Economically speaking, the inequality associated with labor and capital income sources accumulates with the inequality created by other structural coalitions of income sources.

If we take a look at the details of the extra pairwise interactions in Table 9, it appears, as expected, that labor and capital income sources interact negatively with social transfer sources, as they do not benefit the same individuals. However, these negative interactions are not high enough to offset the positive interactions between the two.

²² The Gini index, noted G , for n individuals ordered from the lowest income to the highest income is denoted by: $G = \frac{2}{n} \sum_{i=1}^n (p_i - L_i)$, where p_i and L_i are the respective cumulative share of population and the cumulative share of income.

Table 9 The weighted sum of pairwise interactions between i and S_ℓ

i/S_ℓ	L_1L_2	T_1T_2	K_1K_2	Total
L_1		– 0.0054	0.0109	0.0054
L_2		– 0.0033	0.0065	0.0033
T_1	–		–	–
T_2	– 0.0087		– 0.0087	– 0.0174
K_1	0.0109	– 0.0054		0.0054
K_2	0.0065	– 0.0033		0.0033

“–” indicates a perfect null value, while an empty cell indicates that there is no interaction term

4.2.3 Third decomposition

The third decomposition is proposed in Table 10, based on Proposition 7. The decomposition is similar to the second decomposition, but the interactions are between the secondary sources. In Sect. 3, we show that the second decomposition of the Shapley–Owen value is weaker than the third decomposition.

Furthermore, owing to the third decomposition proposed in this article, it is possible to decompose the interaction of any source with the other sources. Table 8 provides all pairwise interactions. Table 11 shows that the pairwise interactions between L_1 and L_2 and between L_1 and K_2 are positive and relatively high. Thus, L_1 has an amplifier link with these two sources as, on average, these two pairwise interactions create more inequality when they are considered together than alone. By contrast, the interaction between L_1 and T_2 is negative, in the sense that T_2 reduces the inequality created by L_1 .

The interpretation of the third decomposition may differ from the second decomposition as it has a different conception of interaction. For instance, according to the third one, the interaction term of the income source L_2 is only due to its positive interaction with L_1 , the other source in its coalition. However, the second decomposition highlighted that L_2 has a negative interaction with the T coalition and a positive one with the K coalition, which is more straightforward as L_2 benefits the same individuals as the K sources (see Table 9).

One can notice that the interaction between two sources that do not belong to the same structural coalition is not necessarily symmetric. For instance, in Table 11, the interaction between $i = L_1$ and $j = T_2$ is equal to -0.0022 , but the other way round ($i = T_2$ and $j = L_1$), it is equal to -0.0054 . This asymmetry is explained by the fact that the sets of coalitions T' , taken into account to measure the interaction, are not the same in both cases. As pointed out previously, each coalition taken into consideration to compute the interaction between i and j can include all the other sources of the structural coalition of i . Thus, as by definition the structural coalition for two players that do not belong to the same group is not the same, the coalitions used for the interaction computation are different. Therefore, unlike a classic Shapley decomposition, the Shapley–Owen decomposition shows that the influence of a source i on a source j is not necessarily the same as the influence of j on i . For instance, we may say in this given case that the presence of L_1 increases the

Table 10 A revisited Shapley–Owen decomposition—third decomposition

	Labor income		Social transfers		Capital income		Sum
	Wage	Bonus	Family	Housing	Financial	Estate	
Ψ^{Sh}	0.0272	0.0163	–	– 0.0087	0.0991	0.0229	0.1569
	17.36%	10.42%	0.00%	– 5.56%	63.19%	14.58%	100%
PMC	0.0327	0.0196	–	– 0.0261	0.1046	0.0261	0.1569
	20.83%	12.50%	0.00%	– 16.67%	66.67%	16.67%	100.00%
$\text{INT}_{v_{i,j}}^{\text{OW}}$	0.0054	0.0033	–	– 0.0174	0.0054	0.0033	0.00
	3.47%	2.08%	0.00%	– 11.11%	3.47%	2.08%	0.00%

The wage contribution is equal to 0.0272, which represents 17.36% of the total inequality measured by the Gini index (0.156863). All percentages are calculated relatively to the total inequality

Table 11 The weighted sum of pairwise interactions between i and j

ilj	L_1	L_2	T_1	T_2	K_1	K_2	Total
L_1		0.0033	–	– 0.0022	0.0000	0.0044	0.0054
L_2	0.0033		–	– 0.0000	0.0000	– 0.0000	0.0033
T_1	–	–		–	– 0.0000	0.0000	–
T_2	– 0.0054	– 0.0033	–		– 0.0054	– 0.0033	– 0.0174
K_1	0.0022	0.0022	– 0.0000	– 0.0022		0.0033	0.0054
K_2	0.0000	– 0.0000	–	0.0000	0.0033		0.0033

“–” indicates a perfect null value, while “0.0000” indicates a value close to zero. As postulated previously, the pairwise interaction between i and j , in the presence of the other sources, is equal to $\sum_{T \in \text{Coal}[\mathcal{S}^{-i-j}]} \text{INT}_{v_{i,j}}^{\text{OW}}(i, j, T)$

importance of T_2 by 0.0054 points, while the presence of T_2 increases the importance of L_1 by 0.0022 points.

Furthermore, by way of additional information, we compare this revisited Owen decomposition to the revisited Shapley decomposition proposed by Chantreuil et al. (2019). The results of this decomposition are given in Table 12. The PMC of each source of income inequality is the same in all cases; the differences between the two decompositions lie in the interaction term. Indeed, the Shapley decomposition disregards the hierarchical structure of income; it considers each source independently from the structural coalition it is included in. For instance, in the Shapley decomposition, the interaction between the wage source and the family social transfer source can solely be computed in the presence of the financial capital income source. This cannot be the case for the Shapley–Owen decomposition which would consider all the capital income sources in such an example. Therefore, the differences lie in another approach of the interactions between the income sources.

Table 12 A revisited Shapley decomposition

	Labor income		Social transfers		Capital income		Sum
	Wage	Bonus	Family	Housing	Financial	Estate	
φ^{Sh}	0.0292	0.0172	–	– 0.0144	0.1011	0.0237	0.1569
	18.61%	10.97%	0.00%	– 9.17%	64.44%	15.14%	100%
PMC	0.0327	0.0196	–	– 0.0261	0.1046	0.0261	0,1569
	20.83%	12.50%	0.00%	– 16.67%	66.67%	16.67%	100%
$\text{INT}_{v_{l,i}}^{\text{SH}}$	0.0035	0.0024	–	– 0.0118	0.0035	0.0024	0,00
	2.22%	1.53%	0.00%	– 7.50%	2.22%	1.53%	0.00%

Here, the Shapley value $\varphi^{\text{Sh}} := \text{PMC} - \text{INT}_{v_{l,i}}^{\text{SH}}$ measures the importance of each source. For instance, the importance of the wage source according to the Shapley value is equal to 0.0292, which represents 18.61% of the total inequality measured by the Gini index (0.156863)

5 Conclusion

In this article, our main results are the new inequality index decompositions based on the Shapley–Owen value, which are the generalizations of the Shapley decomposition proposed by Chantreuil et al. (2019). These decompositions are especially suited when there is a hierarchical income distribution. In such a case, we show that the importance of an income source can be rewritten as its pure marginal contribution, minus a term of pairwise interactions. Thus, we show why the impact of removing wage differences associated with one source does not match its importance to inequality. The difference between the two is explained by the interaction that a source can have with the other sources. Indeed, the importance of a source of inequality depends on the other sources. Our tool can identify the nature of the link between each pair of sources and compute the intensity of this link. Moreover, even though we apply our new decompositions to the famous Gini index, it can also be used with a wide range of inequality indices. We believe that these new decompositions can be very useful to policymakers in plainly understanding the role of each income source in inequality and the evolution of their importance over time. Therefore, they may help policymakers to appreciate the consequences of an equalizing public policy on inequality.

Appendix: Proofs

Proof of Theorem 1 $\implies$) Following the proof of Proposition 8 given by Albizuri and Zarzuelo (2004), the coalitional value Ψ given by Eq. (2) satisfies the linearity, carrier (ii), rearrangement and coalitional partnership. The conditions

$\sum_{p=0}^{m-1} \binom{m-1}{p} a(m, p) = 1$ and $\sum_{k=0}^{s_{m_i}-1} \binom{s_{m_i}-1}{k} b(s_{m_i}, k) = 1$ guarantee that the coalitional semivalue satisfies the dummy player property.

$\Leftarrow$) The dummy player property is stronger than the projection property. In the proof of Proposition 11 advanced by Albizuri and Zarzuelo (2004), Proposition 11 is necessity and they used Lemma 9. Therefore, the corner stone of their result is Lemma 9. In Proposition 11, when we replace the projection by the dummy player property and maintain their proof, Eq. (28) of Lemma 9 holds without using the monotonicity, since all players outside of N , even R , are null players. Therefore, the dummy player property alone would imply Eq. (28). In this case, when we follow the rest of the proof, the monotonicity property will be useful only to show that the families of real $\rho^{N,t,s}$ are nonnegative. If we dropped the monotonicity property, the coefficients $\rho^{N,t,s}$ would be the real numbers. Consequently, the coefficients $a(m, p)$ and $b(s_{m_i}, k)$ given in Proposition 11 should be the real numbers.

Proof of Proposition 2 This proof will be obtained by induction. For any $N \in \mathcal{N}$, $v, w \in \Gamma(N)$, $i, j \in N$, $i \neq j$, and $T \subseteq N \setminus ij$.

Induction basis: for $t = 0$, we obtain $[v(ij) - v(j)] - [v(j) - v(\emptyset)] = \lambda_{ij}(v)$, and $[w(ij) - w(j)] - [w(j) - w(\emptyset)] = \lambda_{ij}(w)$ at the same time, which proves the claim.

Induction hypothesis: let the result hold for all $T \subseteq N \setminus ij$, such that $0 \leq t \leq k < n - 2$.

Induction step: consider T being such that $t = k + 1$, then we directly obtain

$$[v(T \cup ij) - v(T \cup j)] - [v(T \cup i) - v(T)] = \sum_{S \subseteq T} \lambda_{S \cup ij}(v) = \lambda_{T \cup ij}(v) + \sum_{S \subsetneq T} \lambda_{S \cup ij}(v).$$

Analogously,

$[w(T \cup ij) - w(T \cup j)] - [w(T \cup i) - w(T)] = \lambda_{T \cup ij}(w) + \sum_{S \subsetneq T} \lambda_{S \cup ij}(w)$. In view of the induction hypothesis, this proves the claim.

Proof of Theorem 2 For any $N \in \mathcal{N}$, $(v, \mathfrak{B}) \in G(N)$ and $k \in M$, if a coalitional value Ψ satisfies the quotient game property, then we have $\sum_{i \in S_k} \Psi_i^N(v, \mathfrak{B}^N) = \Psi_k^M(v^q, [M])$. Following Courtin et al. (2020) (Theorem 2), a value is AID^{**} , if and only, it satisfies linearity, anonymity (rearrangement when the mapping π is one-to-one), and dummy player property.

Proof of Corollary 1 We use Theorem 2 stated in this document and Corollary 2 proposed by Courtin et al. (2020).

Proof of Theorem 3 Following Theorem 1, it will suffice for us to prove that a coalitional value is AID^* , if and only if, for any $N \in \mathcal{N}$, $(v, \mathfrak{B}) \in G(N)$ and $i \in N$, there exist two families of **real numbers** $\{a(m, p)\}_{p=0, \dots, m-1}$ and

$\{b(s_{m_i}, k)\}_{k=0, \dots, s_{m_i}-1}$ fulfilling $\sum_{p=0}^{m-1} \binom{m-1}{p} a(m, p) = 1$ and $\sum_{k=0}^{s_{m_i}-1} \binom{s_{m_i}-1}{k} b(s_{m_i}, k) = 1$, such that

$$\Psi_i(v, \mathfrak{B}) := \sum_{P \subseteq (M \setminus m_i)} \sum_{K \subseteq (S_{m_i} \setminus i)} a(m, p) \cdot b(s_{m_i}, k) [v(\mathbf{P}_N \cup K \cup i) - v(\mathbf{P}_N \cup K)].$$

$\Rightarrow$) Let an AID * coalitional value Ψ . From Definition 9, for any $N \in \mathcal{N}$ and $(v, \mathfrak{B}) \in G(N)$, there exist two families of real numbers $\{g(s_{m_i}, k)\}_{k=0}^{s_{m_i}-1}$ and $\{f(m, p)\}_{p=0}^{m-2}$, such that $\sum_{k=0}^{s_{m_i}-1} \binom{s_{m_i}-1}{k} g(s_{m_i}, k) = 1$ and

$$\begin{aligned} \Psi_i(v, \mathfrak{B}) &= \sum_{K \subseteq S_{m_i} \setminus i} g(s_{m_i}, k) [v(i \cup K \cup N \setminus S_{m_i}) - v(K \cup N \setminus S_{m_i})] \\ &\quad - \sum_{\ell \in M \setminus m_i} \sum_{P \subseteq M \setminus \{\ell, m_i\}} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p) \cdot g(s_{m_i}, k) \cdot \overline{\text{Int}}_v^{\mathfrak{B}^N}(i, S_\ell, T) \\ &= \sum_{K \subseteq S_{m_i} \setminus i} g(s_{m_i}, k) [v(i \cup K \cup N \setminus S_{m_i}) - v(K \cup N \setminus S_{m_i})] - [a - b - c + d], \end{aligned} \quad (8)$$

$$\begin{aligned} \text{with } a &:= \sum_{\ell \in M \setminus m_i} \sum_{P \subseteq M \setminus \{\ell, m_i\}} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p) \cdot g(s_{m_i}, k) \cdot v(i \cup S_\ell \cup \mathbf{P}_N \cup K) \\ &= \sum_{\ell \in M \setminus m_i} \sum_{P \subseteq M \setminus m_i} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p-1) \cdot g(s_{m_i}, k) \cdot v(i \cup \mathbf{P}_N \cup K) \\ &\quad \ell \in P \\ &= \sum_{P \subseteq M \setminus m_i} \sum_{\ell \in P} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p-1) \cdot g(s_{m_i}, k) \cdot v(i \cup \mathbf{P}_N \cup K) \\ &\quad P \neq \emptyset \\ &= \sum_{P \subseteq M \setminus m_i} \sum_{K \subseteq S_{m_i} \setminus i} p \times f(m, p-1) \cdot g(s_{m_i}, k) \cdot v(i \cup \mathbf{P}_N \cup K), \\ &\quad P \neq \emptyset \\ b &:= \sum_{\ell \in M \setminus m_i} \sum_{P \subseteq M \setminus \{\ell, m_i\}} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p) \cdot g(s_{m_i}, k) \cdot v(i \cup \mathbf{P}_N \cup K) \\ &= \sum_{\ell \in M \setminus m_i} \sum_{P \subset M \setminus m_i} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p) \cdot g(s_{m_i}, k) \cdot v(i \cup \mathbf{P}_N \cup K) \\ &\quad \ell \notin P \end{aligned}$$

$$\begin{aligned}
&= \sum_{P \subset M \setminus m_i} \sum_{\ell \notin m_i \cup P} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p) \cdot g(s_{m_i}, k) \cdot v(i \cup \mathbf{P}_N \cup K) \\
&= \sum_{P \subset M \setminus m_i} \sum_{K \subseteq S_{m_i} \setminus i} (m - p - 1) f(m, p) \cdot g(s_{m_i}, k) \cdot v(i \cup \mathbf{P}_N \cup K), \\
\text{while } c &:= \sum_{\ell \in M \setminus m_i} \sum_{P \subseteq M \setminus \{\ell, m_i\}} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p) \cdot g(s_{m_i}, k) \cdot v(S_\ell \cup \mathbf{P}_N \cup K) \\
&= \sum_{\substack{P \subseteq M \setminus m_i \\ P \neq \emptyset}} \sum_{K \subseteq S_{m_i} \setminus i} p \cdot f(m, p - 1) \cdot g(s_{m_i}, k) \cdot v(\mathbf{P}_N \cup K) \\
\text{and } d &:= \sum_{\ell \in M \setminus m_i} \sum_{P \subseteq M \setminus \{\ell, m_i\}} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p) \cdot g(s_{m_i}, k) \cdot v(\mathbf{P}_N \cup K) \\
&= \sum_{P \subset M \setminus m_i} \sum_{K \subseteq S_{m_i} \setminus i} (m - p - 1) \cdot f(m, p) \cdot g(s_{m_i}, k) \cdot v(\mathbf{P}_N \cup K).
\end{aligned}$$

Equation (8) becomes

$$\Psi_i(v, \mathfrak{B}) = \sum_{P \subseteq (M \setminus m_i)} \sum_{K \subseteq (S_{m_i} \setminus i)} a(m, p) \cdot b(s_{m_i}, k) [v(\mathbf{P}_N \cup K \cup i) - v(\mathbf{P}_N \cup K)],$$

where $b(s_{m_i}, k) = g(s_{m_i}, k)$, and

$$\begin{aligned}
a(m, p) &= [(m - 1 - p)f(m, p) - pf(m, p - 1)]e(m, p) + (m - 1)f(m, 0) \cdot r(m, p) \\
&\quad + [1 - (m - 1)f(m, m - 2)]\ell(m, p),
\end{aligned}$$

$$\text{with } e(m, p) = \begin{cases} 1 & \text{if } 0 < p < m - 1 \\ 0 & \text{otherwise} \end{cases}; \quad \ell(m, p) = \begin{cases} 1 & \text{if } p = m - 1 \\ 0 & \text{otherwise} \end{cases} \text{ and}$$

$$r(m, p) = \begin{cases} 1 & \text{if } p = 0 \\ 0 & \text{otherwise} \end{cases}.$$

Moreover, $\sum_{p=0}^{m-1} \binom{m-1}{p} a(m, p) = 1$. In fact

$$\begin{aligned}
\sum_{p=0}^{m-1} \binom{m-1}{p} a(m, p) &= (m-1)f(m, 0) + 1 - (m-1)f(m, m-2) \\
&\quad + \sum_{p=1}^{m-2} (m-p-1) \binom{m-1}{p} f(m, p) - \sum_{p=1}^{m-2} p \binom{m-1}{p} f(m, p-1) \\
&= 1 + \sum_{p=0}^{m-2} (m-p-1) \binom{m-1}{p} f(m, p) - \sum_{p=1}^{m-1} p \binom{m-1}{p} f(m, p-1) \\
&= 1 + \sum_{p=0}^{m-2} (m-p-1) \binom{m-1}{p} f(m, p) - \sum_{p=0}^{m-2} (p+1) \binom{m-1}{p+1} f(m, p) \\
&= 1.
\end{aligned}$$

$\Leftarrow$) We suppose that for any $N \in \mathcal{N}$, $(v, \mathfrak{B})$ and $i \in N$, there exist two families of **real numbers** $\{a(m, p)\}_{p=0, \dots, m-1}$ and $\{b(s_{m_i}, k)\}_{k=0, \dots, s_{m_i}-1}$ fulfilling $\sum_{p=0}^{m-1} \binom{m-1}{p} a(m, p) = 1$ and $\sum_{k=0}^{s_{m_i}-1} \binom{s_{m_i}-1}{k} b(s_{m_i}, k) = 1$, such that

$$\Psi_i(v, \mathfrak{B}) := \sum_{P \subseteq (M \setminus m_i)} \sum_{K \subseteq (S_{m_i} \setminus i)} a(m, p) \cdot b(s_{m_i}, k) [v(\mathbf{P}_N \cup K \cup i) - v(\mathbf{P}_N \cup K)].$$

If we set $g(s_{m_i}, k) = b(s_{m_i}, k)$ and

$$f(m, p) = \begin{cases} \frac{a(m, 0)}{m-1} & \text{if } p = 0 \\ \frac{1 - a(m-1)}{m-1} & \text{if } t = m-2 \\ \frac{p \times f(m, p-1) + a(m, p)}{m-p-1} & \text{otherwise} \end{cases},$$

then

$$\begin{aligned}
a(m, p) &= [(m-1-p)f(m, p) - pf(m, p-1)]e(m, p) + (m-1)f(m, 0) \cdot r(m, p) \\
&\quad + [1 - (m-1)f(m, m-2)]\ell(m, p),
\end{aligned}$$

and

$$\begin{aligned}
\Psi_i(v, \mathfrak{B}) &:= \sum_{K \subseteq S_{m_i} \setminus i} g(s_{m_i}, k) [v(i \cup K \cup M \setminus S_{m_i}) - v(K \cup N \setminus S_{m_i})] \\
&\quad - \sum_{\ell \in M \setminus m_i} \sum_{P \subseteq M \setminus \{\ell, m_i\}} \sum_{K \subseteq S_{m_i} \setminus i} f(m, p) \cdot g(s_{m_i}, k) \cdot \overline{\text{Int}}_v^{\mathfrak{B}^N}(i, S_\ell, T).
\end{aligned}$$

Albizuri and Zarzuelo (2004) show that every coalitional semivalue satisfies the linearity property, and we know that by linearity, everything is determined on dirac games ($\delta_T(S) = 1$ if $S = T$ and 0 otherwise).

$$\Psi_i(\delta_N, \mathfrak{B}) = a(m, m-1) \cdot b(s_{m_i}, s_{m_i}-1) = \\ b(s_{m_i}, s_{m_i}-1) - (m-1)b(s_{m_i}, s_{m_i}-1) \cdot f(m, m-2).$$

Then

$$f(m, m-2) = \frac{1 - a(m, m-1)}{(m-1)}.$$

The families of real numbers $\{g(s_{m_i}, k)\}_{k, \dots, s_{m_i}-1}$ exist (coefficients of value φ). It suffices to set

$$f(m, p) = \begin{cases} \frac{a(m, 0)}{m-1} & \text{if } p = 0 \\ \frac{p \times f(m, p-1) + a(m, p)}{m-p-1} & \text{otherwise,} \end{cases}$$

and we conclude that Ψ is AID * .

Proof outline of Proposition 3

- (i) $\sum_{i \in S_\ell} [\Psi_i^{\text{Sh}}(v^N, \mathfrak{B}^N) - \Psi_i^{\text{Sh}}(v^{N \setminus S_r}, \mathfrak{B}^{N \setminus S_r})] = \\ \sum_{i \in S_\ell} \sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus S_r, i}]} \text{INT}_v^{\text{extra}}(i, S_r, T). \quad \sum_{i \in S_r} [\Psi_i^{\text{Sh}}(v^N, \mathfrak{B}^N) - \Psi_i^{\text{Sh}}(v^{N \setminus S_\ell}, \\ \mathfrak{B}^{N \setminus S_\ell})] = \sum_{i \in S_r} \sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus S_\ell, i}]} \text{INT}_v^{\text{extra}}(i, S_\ell, T).$
- (ii) The Shapley value satisfies the Balanced contributions property i.e., for all $v \in \Gamma(N)$, $i, j \in N$, $\varphi_j^{\text{Sh}}(v^N) - \varphi_j^{\text{Sh}}(v^{N \setminus i}) = \varphi_i^{\text{Sh}}(v^N) - \varphi_i^{\text{Sh}}(v^{N \setminus j})$.
- (iii) The Owen value satisfies the *quotient game property* (see Alonso-Mejide et al. 2007).

The statements (ii) and (iii) imply that the Owen value satisfies the Balanced contribution across the structural coalitions property. More formally, that is to say that

$$\sum_{i \in S_\ell} [\Psi_i^{\text{Sh}}(v^N, \mathfrak{B}^N) - \Psi_i^{\text{Sh}}(v^{N \setminus S_r}, \mathfrak{B}^{N \setminus S_r})] = \\ \sum_{i \in S_r} [\Psi_i^{\text{Sh}}(v^N, \mathfrak{B}^N) - \Psi_i^{\text{Sh}}(v^{N \setminus S_\ell}, \mathfrak{B}^{N \setminus S_\ell})].$$

Proof of Theorem 4

First part: Let Ψ be a coalitional value and for all $v \in \Gamma(N)$ and for any $\mathfrak{B}^N := \{S_1, S_2, \dots, S_m\}$ a coalition structure over N . If Ψ satisfies the efficiency, then we have

$$\begin{aligned}
v(N) - v(N \setminus i) &= \sum_{j \in N} \Psi_j^N(v^N, \mathfrak{B}^N) - \sum_{j \in N \setminus i} \Psi_j^N(v^{N \setminus i}, \mathfrak{B}^{N \setminus i}) \\
&= \Psi_i^N(v^N, \mathfrak{B}) + \sum_{j \in S_{m_i} \setminus i} [\Psi_j^N(v^N, \mathfrak{B}^N) - \Psi_j^N(v^{N \setminus i}, \mathfrak{B}^{N \setminus i})] \\
&\quad - \sum_{\ell \in M \setminus m_i} \sum_{j \in S_\ell} [\Psi_j^N(v^N, \mathfrak{B}^N) - \Psi_j^N(v^{N \setminus i}, \mathfrak{B}^{N \setminus i})].
\end{aligned}$$

Moreover, if Ψ satisfies the linearity, rearrangement, projection, and coalitional partnership, then Ψ is the Shapley–Owen value. As a result of lengthy calculations, if $j \in S_{m_i}$, then we have

$$\begin{aligned}
&\Psi_j^N(v^N, \mathfrak{B}^N) - \Psi_j^N(v^{N \setminus i}, \mathfrak{B}^{N \setminus i}) \\
&= \sum_{j \in S_{m_i} \setminus i} \sum_{\substack{T \in \text{Coal}[\mathfrak{B}^N, i] \\ j \notin T}} \frac{p!(m-p-1)!}{m!} \cdot \frac{(k+1)!(s_{m_j}-k-2)!}{s_{m_j}!} \cdot \text{Int}_v^N(i, j, T),
\end{aligned}$$

and if $j \notin S_{m_i}$, then we have

$$\begin{aligned}
&\Psi_j^N(v^N, \mathfrak{B}^N) - \Psi_j^N(v^{N \setminus i}, \mathfrak{B}^{N \setminus i}) \\
&= \sum_{\ell \in M \setminus m_i} \sum_{j \in S_\ell} \sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus S_{m_i}}, i]} \frac{(p+1)!(m-p-2)!}{m!} \times \frac{k!(s_{m_j}-k-1)!}{s_{m_j}!} \cdot \text{Int}_v^N(i, j, T \cup (S_{m_i} \setminus i)).
\end{aligned}$$

Finally, we have

$$\begin{aligned}
&\Psi_i^N(v^N, \mathfrak{B}^N) \\
&= v(N) - v(N \setminus i) - \sum_{j \in S_{m_i} \setminus i} \sum_{\substack{T \in \text{Coal}[\mathfrak{B}^N, i] \\ j \notin T}} \frac{p!(m-p-1)!}{m!} \cdot \frac{(k+1)!(s_{m_j}-k-2)!}{s_{m_j}!} \text{Int}_v^N(i, j, T) \\
&\quad - \sum_{\ell \in M \setminus m_i} \sum_{j \in S_\ell} \sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus S_{m_i}}, i]} \frac{(p+1)!(m-p-2)!}{m!} \times \frac{k!(s_{m_j}-k-1)!}{s_{m_j}!} \text{Int}_v^N(i, j, T \cup (S_{m_i} \setminus i)).
\end{aligned}$$

Second part: Let us show that, $\sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus i}, j]} 2Q_T = 1$.

First case: $Q_T = \frac{(p+1)!(m-p-2)!}{m!} \times \frac{k!(s_{m_j}-k-1)!}{s_{m_j}!}$ if $S_{m_i} \setminus i = \emptyset$.

Thus,

$$\sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus i}, j]} 2Q_T = \sum_{p=0}^{m-2} \sum_{k=0}^{s_{m_j}-1} 2 \binom{m-2}{p} \frac{(p+1)!(m-p-2)!}{m!} \times \binom{s_{m_j}-1}{k} \frac{k!(s_{m_j}-k-1)!}{s_{m_j}!}.$$

It yields that $\sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus i}, j]} 2Q_T = \sum_{p=0}^{m-2} \sum_{k=0}^{s_{m_j}-1} \frac{2(p+1)}{m(m-1)} \times \frac{1}{s_{m_j}} = \sum_{p=0}^{m-2} \frac{p+1}{m(m-1)}$.

Hence, $\sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus i}, j]} 2Q_T = \frac{2}{m(m-1)} \left[\frac{(m-1)(m-2)}{2} + (m-1) \right] = 1$. As desired.

Second case: $Q_T = \frac{p!(m-p-1)!}{m!} \times \frac{(k+1)!(s_{m_j}-k-2)!}{s_{m_j}!}$ if $S_{m_i} = S_{m_j}$. As previously,

$$\sum_{T \in \text{Coal}[\mathfrak{B}^{N \setminus i, j}]} 2Q_T = 1.$$

$$\text{Third case: } Q_T = \begin{cases} \frac{p!(m-p-1)!}{m!} \times \frac{k!(s_{m_j} - k - 1)!}{s_{m_j}!} & \text{if } (S_{m_i} \setminus i) \subset T, \\ 0 & \text{if } (S_{m_i} \setminus i) \subset (N \setminus T) \end{cases},$$

and then,

$$\sum_{\substack{T \in \text{Coal}[\mathfrak{B}^{N \setminus i, j}] \\ (S_{m_i} \setminus i) \subset T}} 2Q_T = 1.$$

Proof outline of Proposition 4 If a coalitional value Ψ satisfies the balanced contributions within a structural coalition property (Alonso-Meijide et al. 2007), i.e., for all $v \in \Gamma(N)$, for any $\mathfrak{B}^N := \{S_1, S_2, \dots, S_m\}$ a coalition structure over N , for all $\ell \in M, i, j \in S_\ell$, $\Psi_j^N(v^N, \mathfrak{B}^N) - \Psi_j^N(v^{N \setminus i}, \mathfrak{B}^{N \setminus i}) = \Psi_i^N(v^N, \mathfrak{B}^N) - \Psi_i^N(v^{N \setminus j}, \mathfrak{B}^{N \setminus j})$. Thus, the result follows, since the Owen value satisfies this property.

Proof of Propositions 5, 6 and 7 Due to the efficiency of the Shapley–Owen value, these propositions are deduced directly from Corollary 2, Corollary 4 and Theorem 4, respectively.

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